Notes on Abstract Space

Zhu Huatao

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These are notes on abstract spaces, mainly based on the book *An Introduction to Abstract Spaces* by Hu Shigeng and Zhang Xianwen.

1 Basic Concepts

1.1 Sets and Relations

We denote by 2^X the power set of X, i.e., the set of all subsets of X. Some related notation is as follows. For any $\mathcal{A} \subset 2^X$, we set

$$\bigcup \mathcal{A} = \bigcup \{A : A \in \mathcal{A}\},$$

and

$$\bigcap \mathcal{A} = \bigcap \{A : A \in \mathcal{A}\}.$$

In addition, we define

$$\mathcal{A}^* = \{\bigcap \mathcal{B} : \mathcal{B} \subset \mathcal{A} \text{ and } |\mathcal{B}| \text{ is finite}\}.$$

If $\mathcal{A}, \mathcal{B} \subset 2^X$, we write

$$\mathcal{A} \vdash \mathcal{B} \iff \forall B \in \mathcal{B},\ \exists A \in \mathcal{A} :\ A \subset B,$$

and

$$\mathcal{A} \prec \mathcal{B} \iff \forall A \in \mathcal{A},\ \exists B \in \mathcal{B} :\ A \subset B.$$

⟨def1.1⟩ **Definition 1.1.** If $\mathcal{A} \subset 2^X$ is nonempty and satisfies

(a) $\emptyset \notin \mathcal{A}$;

(b) For any $A_1, A_2 \in \mathcal{A}$, $A_1 \cap A_2 \in \mathcal{A}$;

(c) If $\mathcal{A} \vdash A \subset X$, then $A \in \mathcal{A}$,

then we call $\mathcal{A}$ a **filter** on X. Moreover, if $\mathcal{B} \subset \mathcal{A}$ satisfies $\mathcal{B} \vdash \mathcal{A}$, then we say that $\mathcal{B}$ is a **base** of the filter $\mathcal{A}$.

It's easy to see that condition (b) in Definition 1.1 can be replaced by $\mathcal{A} = \mathcal{A}^*$.

Any filter has at least one base, for example, itself. Conversely, any nonempty $\mathcal{B} \subset 2^X$ satisfying suitable conditions is a base of a filter.

Theorem 1.1. *If a nonempty $\mathcal{B} \subset 2^X$ satisfies*

(a) $\emptyset \notin \mathcal{B}$;

(b) *For any $B_1, B_2 \in \mathcal{B}$, we have $\mathcal{B} \vdash B_1 \cap B_2$,*

then $\mathcal{B}$ is a base of a filter $\mathcal{A} \subset 2^X$.

Proof. Let

$$\mathcal{A} = \{A \subset X : \mathcal{B} \vdash A\}.$$

It is straightforward to verify that $\mathcal{A}$ is a filter and that $\mathcal{B}$ is a base of $\mathcal{A}$. □

Next, we discuss relations between two nonempty sets. For any two nonempty sets X and Y, we consider elements $x \in X$ and $y \in Y$ as abstract variables, and define how x is related to y.

?⟨def1.3⟩? **Definition 1.2.** Let X and Y be two nonempty sets. Any subset $F \subset X \times Y$ is called a **relation** from X to Y. When $(x, y) \in F$, we say that x is related to y by the relation F, denoted by xFy or $y \in Fx$. If $F \subset X \times X$, we call it a **binary relation** on X.

For any relation F from X to Y and any subset $A \subset X$, define

$$F(A) = \{y \in Y : \exists x \in A,\ (x, y) \in F\}.$$

Let $Fx = F\{x\}$. Then $F(A) = \bigcup_{x \in A} Fx$. If $G \subset Y \times Z$ is another relation, the composition $G \circ F$ is defined by

$$G \circ F = \{(x, z) \in X \times Z : \exists y \in Y,\ (x, y) \in F \text{ and } (y, z) \in G\}.$$

The inverse relation of F is

$$F^{-1} = \{(y, x) \in Y \times X : (x, y) \in F\}.$$

Let us consider any relation $F \subset X \times X$. If $F^{-1} = F$, we say that F is **symmetric**. If $F \circ F \subset F$, we say that F is **transitive**. If the diagonal $\{(x, x) : x \in X\}$ of $X \times X$ is contained in F, we say that F is **reflexive**. If F is symmetric, transitive, and reflexive, we say that F is an **equivalence relation** on X.

Fix a relation $F \subset X \times Y$. We can view F as the correspondence

$$F : X \to 2^Y, \quad x \mapsto Fx.$$

Thus F is a set-valued function. In applications, we often consider the special case that for each $x \in X$, the set Fx contains exactly one element. In that case we say F is a **function** from X to Y.

2 Abstract Space

2.1 Uniform Space

The definition of uniform space is similar to that of topological space. The difference is that in a uniform space, the neighborhoods of points are defined by relations instead of subsets.

⟨def2.1⟩ **Definition 2.1.** Let X be a nonempty set. A nonempty family $\mathcal{U}$ of relations on X is called a **uniform structure** on X if it satisfies the following conditions:

(a) For any $U \in \mathcal{U}$, $\Delta_X \subset U$, where $\Delta_X = \{(x, x) : x \in X\}$ is the diagonal of $X \times X$;

(b) If $U \in \mathcal{U}$, then $U^{-1} \in \mathcal{U}$;

(c) For any $U \in \mathcal{U}$, there exists $V \in \mathcal{U}$ such that $V \circ V \subset U$;

(d) $\mathcal{U}$ is a filter on $X \times X$.

The space X or the pair $(X,\mathcal{U})$ is called a **uniform space**.

For any base of the filter $\mathcal{U}$, we call it a **base** of the uniform structure $(X,\mathcal{U})$. For any $U \in \mathcal{U}$, $U \cap U^{-1} = (U \cap U^{-1})^{-1} \in \mathcal{U}$. Moreover, we have an element $V \in \mathcal{U}$ such that $V \circ V \subset U$. By the fact that $V = V \circ \Delta \subset V \circ V$, we have $V \circ V \in \mathcal{U}$. Thus the set $\{U \cap U^{-1} : U \in \mathcal{U}\}$ or $\{V \circ V : V \in \mathcal{U}\}$ either forms a base of $\mathcal{U}$.

The condition (b) in Definition (2.1) means that for any $x, y \in X$,

$$x = y \iff (x, y) \in U \text{ for all } U \in \mathcal{U}.$$

We can say that the more $U \in \mathcal{U}$ "small", the closer x and y are. The uniform structure $\mathcal{U}$ provides a set of criteria that measure the closeness between two points in the space X. Those facts naturally lead to the definition of uniform neighborhood, i.e.,

$$\mathcal{N}_x = \{U(x) : U \in \mathcal{U}\}, \quad \text{for any } x \in X.$$

Next, we are going to prove that, there exists a unique topology τ on X such that for any $x \in X$, $\mathcal{N}_x$ is the neighborhood system of x with respect to the topology τ. We call τ the **uniform topology** induced by the uniform structure $\mathcal{U}$.

⟨thm2.1⟩ **Theorem 2.1.** *Suppose that $(X,\mathcal{U})$ is a uniform space. Then the following conclusions holds:*

(a) *There exists a unique hausdorff topology τ on X such that for any $x \in X$, $\mathcal{N}_x$ is the neighborhood system of x with respect to τ.*

(b) *Suppose that $\mathcal{B} \subset \mathcal{U}$ is a base of $\mathcal{U}$. Define*

$$\mathcal{B}_x = \{B(x) : B \in \mathcal{B}\}.$$

Then for any $x \in X$, $\mathcal{B}_x$ is a base of the neighborhood system $\mathcal{N}_x$ with respect to the uniform topology τ. Thus if $\mathcal{B}$ is countable, the uniform topology τ is first countable.

(c) *For any $A \in X$ and $M \in X \times X$, we have the following closure formulas with respect to the uniform topology τ:*

$$\bar{A} = \bigcap_{U \in \mathcal{U}} U(A)$$

and

$$\bar{M} = \bigcap_{U \in \mathcal{U}} (U \circ M \circ U).$$

The uniform structure $\mathcal{U}$ can be replaced by any base of $\mathcal{U}$.

Proof. (a) define

$$\tau = \{A \subset X : \forall x \in A,\ \exists U \in \mathcal{N}_x,\ U \subset A\}.$$

First we prove that τ is a topology on X. We have that $X, \emptyset \in \tau$. For any $A, B \in \tau$, we have $A \cap B \in \tau$. In fact, for any $x \in A \cap B$, there exist $U, V \in \mathcal{N}_x$ such that $U \subset A$ and $V \subset B$. Then $U \cap V \in \mathcal{N}_x$ and $U \cap V \subset A \cap B$. Finally, for any family $\{A_i\}_{i \in I} \subset \tau$, we have $\bigcup_{i \in I} A_i \in \tau$. In fact, for any $x \in \bigcup_{i \in I} A_i$, there exists $j \in I$ such that $x \in A_j$. Since $A_j \in \tau$, there exists $U \in \mathcal{N}_x$ such that $U \subset A_j \subset \bigcup_{i \in I} A_i$. Thus τ is a topology on X.

Next we prove that for any $x \in X$, $\mathcal{N}_x$ is the neighborhood system of x with respect to the topology τ. Let $\mathcal{A}_x$ be the neighborhood system of x with respect to τ. For any $U \in \mathcal{N}_x$, by the definition of τ, we have $U \in \mathcal{A}_x$ and thus $\mathcal{N}_x \subset \mathcal{A}_x$. Conversely, for any $A \in \mathcal{A}_x$, by the definition of τ, there exists $U \in \mathcal{N}_x$ such that $U \subset A$. Thus we have $A \in \mathcal{N}_x$ and $\mathcal{A}_x \subset \mathcal{N}_x$. Therefore, $\mathcal{N}_x$ is the neighborhood system of x with respect to the topology τ.

Finally, we prove that the topology τ is hausdorff. For any two distinct points $x, y \in X$, there exists $U \in \mathcal{U}$ such that $(x, y) \notin U$. Then $y \notin U(x)$. Since $U(x) \in \mathcal{N}_x$, then $U(x)$ is a neighborhood of x not containing y. Similarly, there exists a neighborhood of y not containing x.

(b) It's straightforward by conclusion (a) of Theorem (2.1).

(c) By the definition of closure, we have

$$x \in \bar{A} \iff \forall U \in \mathcal{U},\ U(x) \cap A \neq \emptyset,$$

which is equivalent to

$$\bar{A} = \bigcap_{U \in \mathcal{U}} U(A).$$

The second closure formula can be proved similarly.

□

From the construction of the neighborhood system $\mathcal{N}_x$, we can see that the local structure of a uniform space is uniform everywhere. In other words, for any two points $x, y \in X$, there exists a natural correspondence between the neighborhood systems $\mathcal{N}_x$ and $\mathcal{N}_y$.

As we have mentionded that the uniform structure $\mathcal{U}$ provides a set of criteria that measure the closeness between two points in the space X, we hope to use some numerical values to describe the closeness. Basicaly, we expect that there exists a metric that induces the uniform structure $\mathcal{U}$. However, this is not always possible.

⟨thm2.2⟩ **Theorem 2.2.** *Let $(X,\mathcal{U})$ be a uniform space. There exists a metric $d(\cdot,\cdot)$ on X inducing the uniform structure $\mathcal{U}$ if and only if $\mathcal{U}$ has a countable base.*

Proof. The only if part is trivial. We only prove the if part. Suppose that $\mathcal{U}$ has a countable base $\{U_n : n \in \mathbb{N}\}$. Without loss of generality, we can assume that U_n is symmetric and

$$U_n \circ U_n \circ U_n \subset U_{n-1} \quad \text{for all } n \geq 1,$$

where we set $U_0 = X \times X$. For any $x, y \in X$, define

$$f(x,y) = \begin{cases} 2^{-n}, & \text{if } (x,y) \in U_n \text{ and } (x,y) \notin U_{n+1} \text{ for some } n \in \mathbb{N}, \\ 0, & \text{if } x = y, \end{cases}$$

and

$$d(x,y) = \inf \left\{ \sum_{i=1}^{m} f(x_{i-1}, x_i) : m \in \mathbb{N}, x_0 = x, x_m = y, x_i \in X \text{ for } i = 1, \ldots, m-1 \right\}.$$

For simplicity, we denote the sum in the above formula by $S(x,y)$. We first prove that d is a metric on X. By the symmetric property of U_n, we know that f and d are symmetric. It's obvious that $d(x,y) = 0$ is equivalent to $x = y$. For any $x, y, z \in X$, we have

$$\begin{aligned} d(x,y) &= \inf S(x,y) \\ &\leq \inf(S(x,z) + S(z,y)) \\ &= d(x,z) + d(z,y). \end{aligned}$$

Thus d is a metric on X. Then we prove that the metric d induces the uniform structure $\mathcal{U}$. Define

$$V_r = \{(x,y) \in X \times X : d(x,y) < r\}, \quad \text{for any } r > 0.$$

We only need to prove that

$$V_{1/2^{n+1}} \subset U_n \subset V_{1/2^{n-1}} \text{ for all } n \in \mathbb{N}.$$

By the definition of d, it's straightforward to see that $U_n \subset V_{1/2^{n-1}}$. Then we prove that $f(x,y) \leq 2d(x,y)$. We fix a finite sequence $\{x_i\}_{i=0}^m$ with $x_0 = x$ and $x_m = y$ and let $a = \sum_{i=1}^m f(x_{i-1}, x_i)$. We are going to show that $f(x,y) \leq 2a$ by induction on n. We choose $m \in \mathbb{N}$ such that $2^{-m} \leq a < 2^{-m+1}$. There exists the largest integer k such that $\sum_{i=1}^{k-1} f(x_{i-1}, x_i) \leq \frac{a}{2}$. Thus we know

$$\sum_{i=k+1}^{m} f(x_{i-1}, x_i) = a - \sum_{i=1}^{k} f(x_{i-1}, x_i) < \frac{a}{2}.$$

By the induction hypothesis, we have

$$f(x, x_{k-1}) \leq 2 \sum_{i=1}^{k-1} f(x_{i-1}, x_i) \leq a,$$

and

$$f(x_k, y) \leq 2 \sum_{i=k+1}^{m} f(x_{i-1}, x_i) < a.$$

It's trivial that $f(x_{k-1}, x_k) \leq a$. Thus $(x, y) \in U_m \circ U_m \circ U_m \subset U_{m-1}$, which implies that $f(x, y) \leq 2^{-m+1} < 2a$. $\square$

Theorem (2.2) shows that a uniform space $(X, \mathcal{U})$ cannot be metrizable in general. However, we can use a function similar to a metric to describe the closeness between two points in X. This leads to the following definition of pseudometric.

?⟨def2.2⟩? **Definition 2.2.** Let X be a nonempty set. A function $d : X \times X \to \mathbb{R}_+$ is called a **pseudometric** on X if it satisfies the following conditions:

(a) For any $x \in X$, $d(x, x) = 0$;

(b) For any $x, y \in X$, $d(x, y) = d(y, x)$;

(c) For any $x, y, z \in X$, $d(x, z) \leq d(x, y) + d(y, z)$.

The pair (X, d) is called a **pseudometric space**.

It's possible that $\mathcal{U}$ can be induced by a family of pseudometrics.

⟨thm2.3⟩ **Theorem 2.3.** *Suppose that $(X, \mathcal{U})$ is a uniform space. Let P be the set of all uniformly continous pseudometrics on X. Then the uniform structure $\mathcal{U}$ is induced by the family P. Moreover, P is the largest family of pseudometrics on X inducing the uniform structure $\mathcal{U}$.*

We call the family P mentioned in Theorem (2.3) a **gage**.

Proof. Since $d = 0$ is an element of P, P is nonempty. We write $\mathcal{U}_P$ for the uniform structure induced by P and will prove that $\mathcal{U} = \mathcal{U}_P$. For any $d \in P$ and $r > 0$, define

$$V_{d,r} = \{(x, y) \in X \times X : d(x, y) < r\}.$$

By the uniform continuity of d, we have $V_{d,r} \in \mathcal{U}$. In addition, we know that $\mathcal{U}_\mathcal{P}$ is the filter generated by the set

$$\mathcal{A} = \{V_{d,r} : d \in P, r > 0\}.$$

Thus we have $\mathcal{U}_P \subset \mathcal{U}$. To prove the converse inclusion, we only need to prove $\mathcal{A} \vdash \mathcal{U}$. For any $U \in \mathcal{U}$, choose $U_1 = U \cap U^{-1}$. There exists a sequence $\{U_n\}$ such that U_n is symmetric and

$$U_n \circ U_n \circ U_n \subset U_{n-1}, \quad n \geq 2,$$

where we set $U_0 = X \times X$. The same as the proof of Theorem (2.2), there exists a pseudometric d on X such that

$$V_{d,1/2^{n+1}} \subset U_n \subset V_{d,1/2^{n-1}} \text{ for all } n \in \mathbb{N}.$$

This also implies that d is uniformly continuous. Thus $d \in P$ and $\mathcal{A} \vdash U$. $\square$

An interesting question is that, given a topological space (X, τ), does there exist a uniform structure $\mathcal{U}$ on X such that τ is the uniform topology induced by $\mathcal{U}$? We can answer the question by using Theorem (2.3).

?⟨cor2.4⟩? **Corollary 2.1.** *Let (X, τ) be a topological space. Then there exists a uniform structure $\mathcal{U}$ on X such that τ is the uniform topology induced by $\mathcal{U}$ if and only if (X, τ) is completely regular.*

Proof. (a) Suppose that τ is induced by a uniform structure $\mathcal{U}$ and P is the gage of $(X,\mathcal{U})$. Fixing $x \in X$ and $d \in P$, define $f_{d,x}(y) = d(x, y)$ for all $y \in X$. Then $f_{d,x}$ is continuous on X and

$$V_{d,r} = \{(y \in X : f_{d,x}(y) < r)\}.$$

The family $\{V_{d,r}\}$ forms a base of topology τ, which implies that (X, τ) is completely regular.

(b) Suppose that (X, τ) is completely regular. There exists a topological embedding F from (X, τ) into a product space $Y = \prod_{\alpha \in I}[0, 1]_\alpha$, where I is a nonempty index set. Then FX is a uniform space. We denote its uniform structure by $\mathcal{V}$. Thus $\mathcal{U} = F^{-1}(\mathcal{V})$ is a uniform structure on X and

$$F : (X,\mathcal{U}) \to (FX, \mathcal{V})$$

is a uniform isomorphism. We denote by τ' the uniform topology induced by $\mathcal{U}$. Then $F : (X, \tau') \to (FX, \tau'')$ is a homeomorphism, where τ'' is the uniform topology induced by $\mathcal{V}$. Thus $\tau = \tau'$, which means that τ is a uniform topology.

□

Finally, we will talk about the existence of limits in a uniform space. It depends on compactness and completeness of the uniform space, while compactness is correspondent to topology and completeness is correspondent to uniform structure. So we focus on the completeness of a uniform space.

⟨def2.3⟩ **Definition 2.3.** Let $(X,\mathcal{U})$ be a uniform space. If a net $\{x_\alpha\}$ in X satisfies that for any $U \in \mathcal{U}$, there exists α_0 such that $(x_\alpha, x_\beta) \in U$ for all $\alpha, \beta \geq \alpha_0$, then we say that $\{x_\alpha\}$ is a **Cauchy net**. If any Cauchy net in X converges to a point in X, then we say that the uniform space $(X,\mathcal{U})$ is **complete**.

One can observe that $\mathcal{U}$ can be replaced by its base in the definition. From Definition (2.3), we can easily verify that X is complete if and only if every Cauchy net has a cluster point in X. This fact is useful in proving the following theorem.

?⟨thm2.5⟩? **Theorem 2.4.** *Let $(X,\mathcal{U})$ be a uniform space. Then X is complete if and only if for any family $\mathcal{A}$ of closed subsets of X with $\mathcal{A}^* \neq \emptyset$, satisfying that if*

$$\forall U \in \mathcal{U}, \exists A \in \mathcal{A}, A \times A \subset U \tag{2.1}$$ eq2.1

then $\bigcap \mathcal{A} \neq \emptyset$.

Proof. Assume that X is complete and $\mathcal{A}$ is a family of closed subsets of X satisfying (2.1). Without loss of generality, we can assume that $\mathcal{A} = \mathcal{A}^*$. For any $A \in \mathcal{A}$, we choose $x_A \in A$ and let $\{x_A\}$ be the net indexed by $\mathcal{A}$. We are going to show that $\{x_A\}$ is a Cauchy net. For any $U \in \mathcal{U}$, there exists $A_0 \in \mathcal{A}$ such that $A_0 \times A_0 \subset U$. For any $A, B \in \mathcal{A}$ with $A, B \subset A_0$, we have $(x_A, x_B) \in A_0 \times A_0 \subset U$. Thus $\{x_A\}$ is a Cauchy net and there exists $x \in X$ such that $x_A \to x$, which implies that $x \in \bigcap \mathcal{A}$ and thus $\bigcap \mathcal{A} \neq \emptyset$.

Conversely, assume that for any family $\mathcal{A}$ of closed subsets of X with $\mathcal{A}^* \neq \emptyset$, satisfying (2.1) then $\bigcap \mathcal{A} \neq \emptyset$. Let $\{x_t\}$ be a Cauchy net in X and set $A_t = \{x_s : s \geq t\}$. Thus $\mathcal{A} = \{\overline{A_t} : t \in T\}$ is a family of closed subsets of X with $\mathcal{A}^* \neq \emptyset$. For any $U \in \mathcal{U}$, there exists $V \in \mathcal{U}$ such that $V \circ V \subset U$. Then there exists t_0 such that $A_{t_0} \times A_{t_0} \subset V$, which implies that for any $t \geq t_0$, we have

$$\overline{A_t} \times \overline{A_t} \subset V \circ V \subset U$$

. Therefore, $\mathcal{A}$ satisfies (2.1) and thus $\bigcap \mathcal{A} \neq \emptyset$. Thus $\{x_t\}$ converges to some point in X, which means that X is complete. □

3 Topological Vector Space

?⟨def3.1⟩? **Definition 3.1.** Let X be a vector space over a field $\mathbb{K}$. A uniform structure $\mathcal{U}$ on X is called a **vector uniform structure** if the addition and scalar multiplication are uniformly continuous with respect to $\mathcal{U}$. The pair $(X,\mathcal{U})$ is called a **topological vector space**.

In the topological vector space (TVS), the vector space structure and the topology structure are compatible with each other. A simple but important fact is that the scalar multiplication and translation are homeomorphisms in a TVS. Thus a TVS is homogeneous, which means that the local structure of a TVS is uniform everywhere. In particular, the neighborhood system of any point in a TVS can be obtained by translating the neighborhood system $\mathcal{N}_0$ of the zero vector. In fact we have

$$\mathcal{N}_x = x + \mathcal{N}_0 = \{x + U : U \in \mathcal{N}_0\} \text{ for any } x \in X.$$

Similarly, when we talk about the limits of sequences or nets in a TVS, we can always translate the sequence or net by a fixed vector and thus only need to consider the limits of sequences or nets converging to the zero vector. Now let us focus on the uniform structure of a TVS. For any $A \subset X$, we define

$$V_A = \{(x, y) \in X \times X : y - x \in A\}$$

We can easily verify that $\mathcal{V} = V_A : A \subset X$ is a uniform structure base on X. We call the uniform structure $\mathcal{W}$ generated by $\mathcal{V}$ the **vector uniform structure** on X. Moreover, we have

$$V_A(x) = x + A \text{ for any } x \in X,\ A \in \mathcal{N}_0.$$

Thus the uniform topology induced by the vector uniform structure $\mathcal{W}$ is the same as the topology generated by the neighborhood system $\mathcal{N}_0$. In other words, the vector uniform structure $\mathcal{W}$ induces the topology of the TVS.

Assume that X, Y are two TVSs over a field $\mathbb{K}$. We denote all the linear continuous functions from X to Y by $\mathcal{L}(X, Y)$. If $T \in \mathcal{L}(X, Y)$ is a homeomorphism and $T^{-1} \in \mathcal{L}(Y, X)$, then we say that T is a **topological isomorphism** from X to Y. For any linear funcion $T \in \mathcal{L}(X, Y)$, we have

$$Tx = Tx_0 + T(x - x_0).$$

Thus the continuity of T at any point $x_0 \in X$ is equivalent to the continuity of T at the zero vector. In addition, the continuity of T at the zero vector is equivalent to the existence of a neighborhood U of the zero vector such that $T(U)$ is a neighborhood of the zero vector in Y. This fact implies that $T \in \mathcal{L}(X, Y)$ if and only if T is uniformly continuous with respect to the vector uniform structures on X and Y. Thus a topological isomorphism must be a uniform isomorphism.

From above we can see that the structure of a TVS is totally determined by its neighborhood system. Thus it's important to study the neighborhood system of a TVS.

⟨thm3.1⟩ **Theorem 3.1.** *Suppose that X is a TVS over a field $\mathbb{K}$ and $\mathcal{B}$ is a neighborhood base of X, then we have the following conclusions:*

(a) *For any $A, B \in \mathcal{B}$, there exists $C \in \mathcal{B}$ such that $C \subset A \cap B$.*

(b) $\bigcap_{A \in \mathcal{B}} A = \{0\}$.

(c) *For any $A \in \mathcal{B}$, there exists $B \in \mathcal{B}$ such that $B + B \subset A$.*

(d) *For any $A \in \mathcal{B}$ and $\lambda \in \mathbb{K}$ with $|\lambda| \leq 1$, there exists $B \in \mathcal{B}$ such that $\lambda B \subset A$.*

(e) *For any $A \in \mathcal{B}$, we have $X = \bigcup_{\lambda \in \mathbb{K}} \lambda A$.*

Conversely, if $\mathcal{B}$ is a family of subsets of X satisfying the above conditions, then there exists a unique TVS structure on X such that $\mathcal{B}$ is a neighborhood base of the zero vector.

Proof. The only if part is straightforward by the definition of a TVS. We only prove the if part. Define $\mathcal{N}_0 = \{A \subset X : \exists B \in \mathcal{B}, B \subset A\}$ and $\mathcal{N}_x = x + \mathcal{N}_0$ for any $x \in X$. Then $\mathcal{N}_x$ is a neighborhood system of x for any $x \in X$. We are going to show that there exists a unique uniform structure $\mathcal{U}$ on X such that for any $x \in X$, $\mathcal{N}_x$ is the neighborhood system of x with respect to the uniform topology induced by $\mathcal{U}$. Define

$$\tau = \{A \subset X : \forall x \in A, \exists U \in \mathcal{N}_x, U \subset A\}.$$

It's straightforward to verify that τ is a topology on X and for any $x \in X$, $\mathcal{N}_x$ is the neighborhood system of x with respect to τ. Moreover, we have $\bigcap_{A \in \mathcal{B}} A = \{0\}$, which implies that τ is hausdorff. Thus there exists a unique uniform structure $\mathcal{U}$ on X such that for any $x \in X$, $\mathcal{N}_x$ is the neighborhood system of x with respect to the uniform topology induced by $\mathcal{U}$.

Finally, we need to verify that the addition and scalar multiplication are uniformly continuous with respect to the uniform structure $\mathcal{U}$. For any $A, B, C, D \in \mathcal{B}$, we have

$$(A + B) + (C + D) = (A + C) + (B + D) \subset A + B + C + D.$$

Thus there exists $E \in \mathcal{B}$ such that $(E + E) + (E + E) = 4E \subset A + B + C + D$, which implies that

$$(E + E) + (E + E) = (E + E) + (E + E) \subset (A + B) + (C + D).$$

This shows that the addition is uniformly continuous with respect to $\mathcal{U}$. For any $A \in \mathcal{B}$ and $\lambda \in \mathbb{K}$ with $|\lambda| \leq 1$, there exists $B \in \mathcal{B}$ such that $\lambda B \subset A$. Thus for any $x, y \in X$ with $y - x \in B$, we have $y - x \in \lambda B$ for some $\lambda \in \mathbb{K}$ with $|\lambda| \leq 1$, which implies that $y - x \in A$. This shows that the scalar multiplication is uniformly continuous with respect to $\mathcal{U}$. □

Theorem (3.1) tells us that when we want to construct a TVS structure on a vector space, we only need to construct a family of subsets satisfying the conditions in Theorem (3.1), instead of constructing the topology and verifying the continuity of addition and scalar multiplication. Thus we can choose some special neighborhood base to simplify our analysis on a TVS.

Proposition 3.1. *The following families of subsets of a vector space X can be used as neighborhood bases of the zero vector in a TVS structure on X:*

(a) *The family of all balanced* 0*-neighborhoods of X.*

(b) *The family of all symmetric* 0*-neighborhoods of X.*

(c) *The family of all closed (or open)* 0*-neighborhoods of X.*

(d) *The family of all closed (or open) balanced* 0*-neighborhoods of X.*

Proof. The proof is straightforward by verifying that each of the families satisfies the conditions in Theorem (3.1). We only verify the fourth one. For any 0-neighborhood A, there exists a closed 0-neighborhood B such that $B \subset A$. Choose a 0-neighborhood C such that $\lambda C \subset B$ for any $|\lambda| \leq 1$, and let $D = \cap_{|\lambda|\leq 1} \lambda C$. Then $\bar{D}$ is a closed balanced 0-neighborhood and $\bar{D} \subset B \subset A$. □

This shows that we can only consider balanced 0-neighborhoods when we want to construct a TVS structure on a vector space. Moreover, from the condition (3) in Theorem (3.1), we can only consider the 0-neighborhoods in the form of $A + A$.

An interesting question is when a TVS is metrizable. From Theorem (2.2), we know that a TVS is metrizable if and only if its vector uniform structure has a countable base. The following theorem gives us a more explicit criterion for the metrizability of a TVS.

?⟨thm3.2⟩? **Theorem 3.2.** *For a TVS X, the following conclusions are equivalent:*

(a) *X is metrizable.*

(b) *X has a countable neighborhood base at the zero vector.*

(c) *X is a F^* space.*

(d) *The topology of X can be induced by a translation invariant metric.*

Proof. We know that (d) implies (a) and (a) implies (b) by Theorem (2.2). We only need to prove that (b) implies (c) and (c) implies (d).

(c) ⇒ (d): Suppose that X has the quasi norm $\|\cdot\|$. Define $d(x, y) = \|x - y\|$ for any $x, y \in X$. Then d is a translation invariant metric on X and induces the topology of X. In fact, for any $x, y, z \in X$, we have

$$\begin{aligned} d(x, y) &= \|x - y\| \\ &\leq K(\|x - z\| + \|z - y\|) \\ &= K(d(x, z) + d(z, y)). \end{aligned}$$

Thus d is a metric on X. trivially, we can notice that d is translation invariant and induces the topology of X. By the definition of a F^* space, we have that d is a translation invariant metric on X inducing the topology of X.

(b) $\Rightarrow$ (c): Suppose that X has a countable neighborhood base $\{A_n : n \in \mathbb{N}\}$ at the zero vector. Without loss of generality, we can assume that A_n is balanced and $A_{n+1}+A_{n+1} \subset A_n$ for all $n \in \mathbb{N}$. We fix a number $r = 0.r_1r_2\cdots r_n$ with $r_i = 0$ or 1 and $r_n = 1$. Define

$$A_k = \sum_{i=1}^{k} r_i A_i$$

with $A_0 = X$ and

$$\|x\| = \inf\{r : x \in A_r\}$$

It is straightforward to verify that $\|\cdot\|$ is a quasi norm on X and thus X is a F^* space. We are going to verify that $\|\cdot\|$ induces the topology of X. We will prove that for any sequence $\{x_n\}$ in X, $x_n \to 0$ if and only if $\|x_n\| \to 0$. Define $V_r = \{x \in X : \|x\| < r\}$. Let j be the smallest integer such that $r_j = 1$. Then we have

$$A_n \subset V_r \subset A_{j-1}$$

So we have that $\|\cdot\|$ induces the topology of X. □

Next, we are going to talk about the operator between two TVSs. Specifically, we are interested in the bounded linear operator between two TVSs. A bounder linear operator is a linear operator that maps bounded sets to bounded sets, which is a natural generalization of a continuous linear operator between two normed spaces.

Theorem 3.3. *Let X, Y be two TVSs and $T : X \to Y$ be a linear operator. If T is continous, then T is bounded. Conversely, if X is metrizable and T is bounded, then T is continuous. If $dim(X) < \infty$, then every linear operator $T : X \to Y$ is bounded.*

Proof. The first conclusion is straightforward by the definition of continuity. For the second conclusion, we assume that X is metrizable and T is bounded but not continuous. Then there exists a sequence $\{x_n\}$ in X such that $x_n \to 0$ but $T(x_n)$ does not converge to 0 in Y. Since T is bounded, there exists a neighborhood V of the zero vector in Y such that for all n, we have $T(x_n) \notin V$. This contradicts the assumption that T is bounded. Therefore, if X is metrizable and T is bounded, then T is continuous.

For the third conclusion, if $\dim(X) < \infty$, then every linear operator from X to any TVS is bounded. □

Sometimes when we want define a linear operator between two TVSs, we only need to define the operator on a dense subset of the domain space.

Theorem 3.4. (a) *Let X, Y be two metrizable TVSs and Y is sequentially complete. Assume that D is a dense subspace of X and $T \in \mathcal{L}(D, Y)$. There exists a unique linear operator $\tilde{T} \in \mathcal{L}(X, Y)$ such that $\tilde{T}|_D = T$.*

(b) *Let X, Y be two B^* spaces and Y is complete. Assume that A is a subset of X such that span(A) is dense in X and $T \in \mathcal{L}(A, Y)$ satisfying*

$$\|\sum_i \alpha_i T x_i\| \leq \beta \sum_i \|\alpha_i x_i\|$$

for all finite sums $\sum_i \alpha_i x_i$ with $x_i \in A$. Then there exists a linear operator $\tilde{T} \in \mathcal{L}(X, Y)$ such that $\|\tilde{T}\| \leq \beta$.

Proof. (a) For any $x \in X$, there exists a sequence $\{x_n\}$ in D such that $x_n \to x$. So $T(x_n)$ is a Cauchy sequence in Y and thus converges to some point in Y, which we denote by $\tilde{T}(x)$. We are going to verify that $\tilde{T}$ is a linear operator and $\tilde{T}|_D = T$. For any $x, y \in X$ and $\alpha \in \mathbb{K}$, we have

$$\begin{aligned}
\tilde{T}(x+y) &= \lim_{n\to\infty} T(x_n + y_n) \\
&= \lim_{n\to\infty} (T(x_n) + T(y_n)) \\
&= \tilde{T}(x) + \tilde{T}(y),
\end{aligned}$$

and

$$\begin{aligned}
\tilde{T}(\alpha x) &= \lim_{n\to\infty} T(\alpha x_n) \\
&= \lim_{n\to\infty} \alpha T(x_n) \\
&= \alpha \tilde{T}(x),
\end{aligned}$$

So $\tilde{T}$ is a linear operator. For any $x \in D$, we have $x_n = x$ for all n, so $\tilde{T}(x) = T(x)$. Therefore, $\tilde{T}|_D = T$. Finally, we need to verify that $\tilde{T}$ is continuous. For any $y \in Y$ and $\epsilon > 0$, there exists a neighborhood V of the zero vector in Y such that $y + V \subset B(y, \epsilon)$. Since T is continuous, there exists a neighborhood U of the zero vector in X such that $T(U) \subset V$. For any $x \in U$, we have $\tilde{T}(x) \in T(U) \subset V$, which implies that $\tilde{T}(x) \in y + V \subset B(y, \epsilon)$. Thus $\tilde{T}$ is continuous and thus $\tilde{T} \in \mathcal{L}(X, Y)$.

(b) Let $D = span(A)$. For any finite sum $\sum_i \alpha_i x_i$ with $x_i \in A$, we denote $x = \sum_i \alpha_i x_i$ and

$$T(x) = \sum_i \alpha_i T x_i$$

. If $x = \sum_i \alpha_i x_i = \sum_i \alpha_i^{'} x_i$, then we have

$$\| \sum_i (\alpha_i - \alpha_i^{'}) T x_i \| \leq \beta \| \sum_i (\alpha_i - \alpha_i^{'}) x_i \| = 0,$$

which implies that $T(x)$ is well defined. It is straightforward by the condition that $T \in \mathcal{L}(D, Y)$ and $\|T\| \leq \beta$. From (a), there exists a linear operator $\tilde{T} \in \mathcal{L}(X, Y)$ such that $\tilde{T}|_D = T$. For any $x \in X$, there exists a sequence $\{x_n\}$ in D such that $x_n \to x$. Thus we have

$$\begin{aligned} \|\tilde{T}(x)\| &= \lim_{n\to\infty} \|\tilde{T}(x_n)\| \\ &= \lim_{n\to\infty} \|T(x_n)\| \\ &\leq \beta \lim_{n\to\infty} \|x_n\| = \beta \|x\|. \end{aligned}$$

Therefore, $\tilde{T} \in \mathcal{L}(X, Y)$ and $\|\tilde{T}\| \leq \beta$.

□

Next we are going to talk about the open mapping theorem and its applications in TVS.

Theorem 3.5 (Open Mapping Theorem)**.** *Let X be a Frechet space and Y be a TVS. If $T \in \mathcal{L}(X, Y)$ and $R(T)$ is second category in Y, then T is an open mapping and $R(T) = Y$, thus Y is also a Frechet space.*

Proof. Assume that $\|\cdot\|$ is the quasi norm on X and $B_r = \{x \in X : \|x\| < r\}$. We only need to verify that for any $r > 0$, $T(B_r)$ is a neighborhood of the zero vector in Y. Since $R(T)$ is second category in Y and

$$R(T) = T(\bigcup_n nB_r) = \bigcup_n nT(B_r),$$

we have that $T(B_r)$ is not a nonwhere dense set in Y. Thus there exists a balanced neighborhood V_r of the zero vector in Y and $b \in Y$ such taht $V_r + b \subset \overline{T(B_r)}$. So we have

$$V_r \subset \overline{T(B_r)} - \overline{T(B_r)} \subset \overline{T(B_{2r})}$$

If we can prove that $\overline{T(B_{r/2})} \subset T(B_r)$, we will have

$$V_{r/4} \subset \overline{T(B_{r/2})} \subset T(B_r),$$

which implies that $T(B_r)$ is a neighborhood of the zero vector in Y. For any $y \in \overline{T(B_{r/2})}$, there exists a sequence $\{x_n\}$ in $B_{r/2}$ such that $T(x_n) \to y$. Since $B_{r/2}$ is a complete metric space, there exists $x \in B_{r/2}$ such that $x_n \to x$. Thus we have $T(x) = y$ and thus $y \in T(B_r)$. So we have that T is an open mapping.

Since $R(T)$ is a open subspace of Y, we have $R(T) = Y$ and thus Y is also a Frechet space. □

Open mapping theorem is a powerful tool in the analysis of TVS. The following facts are direct applications of the open mapping theorem.

⟨cor3.1⟩ **Corollary 3.1.** *Let X, Y be two Frechet spaces and $T \in \mathcal{L}(X, Y)$. If T is injective, then T is a topological isomorphism from X to $R(T)$. If T is surjective, then T is a topological isomorphism from $X/ker(T)$ to Y.*

(a) *If T is surjective, then T is an open mapping. The mapping $T : X \to R(T)$ is open if and only if $R(T)$ is d in Y.*

(b) *If T is bijective, then T^{-1} is continous and T is a topological isomorphism from X to Y.*

(c) *If (X, τ) and (X, τ_1) are two Frechet spaces and $\tau_1 \subset \tau$, then we have $\tau_1 = \tau$.*

From (b) in Corollary (3.1), we can see that the continuity of T and completeness of X implies the continuity of T^{-1}.

Theorem 3.6 (Closed Graph Theorem)**.** *Let X, Y be two Frechet spaces and $T : X \to Y$ be a linear operator. If the graph of T*

$$G(T) = \{(x, T(x)) : x \in X\}$$

is closed in $X \times Y$, then T is continuous.

Proof. We only need to verify that T is continuous at the zero vector. For any $y \in Y$ and $\epsilon > 0$, there exists a neighborhood V of the zero vector in Y such that $y + V \subset B(y, \epsilon)$. Since $G(T)$ is closed in $X \times Y$, we have that $G(T) \cap (X \times (y + V))$ is closed in $X \times (y + V)$. Since $G(T) \cap (X \times (y + V))$ is not empty, there exists a neighborhood U of the zero vector in X such that

$$(U \times (y + V)) \cap G(T) = (U \times (y + V)) \cap G(T) \cap (X \times (y + V)) = U \times (y + V).$$

Thus for any $x \in U$, we have

$$(x, T(x)) \in G(T) \cap (X \times (y + V)),$$

which implies that

$$T(x) \in y + V \subset B(y, \epsilon).$$

Therefore, T is continuous at the zero vector and thus continuous on X. $\square$

The closed graph theorem is a powerful tool when we want to verify the continuity of a linear operator between two TVSs. For example, if we want to verify the continuity of a linear operator $T : X \to Y$, we only need to verify that for any sequence $\{x_n\}$ in X with $x_n \to 0$ and $T(x_n) \to y$, we have $y = 0$.

Sometimes the continuity of a linear operator is not enough for our analysis. We expect the linear operator to have some kind of uniform behavior.

Definition 3.2. Let $F \subset \mathcal{L}(X, Y)$ be a family of linear operators from X to Y. For $A \subset X$, if $F(A)$ is bounded in Y, then we say that F is **equibounded** at A. If F is equibounded at every bounded subset of X, then we say that F is **equibounded**. For any 0-neighborhood $V \subset Y$, if there exists a neighborhood U of the zero vector in X such that $F(U) \subset V$, then we say that F is **equicontinuous**.

Here are some basic properties of equibounded and equicontinuous families of linear operators.

⟨prop3.2⟩ **Proposition 3.2.** *For $F \subset \mathcal{L}(X, Y)$ and the following conditions:*

(a) *F is equibounded at some neighborhood of the zero vector in X.*

(b) *F is equicontinuous.*

(c) *F is equibounded..*

We have that (1) implies (2) and (2) implies (3). If X, Y are B^* spaces. then

$$(1) \Leftrightarrow (2) \Leftrightarrow (3) \Leftrightarrow \sup_{T \in F} \|T\| < \infty$$

Proof. It is straightforward to verify that (1) implies (2). Now we verify that (2) implies (3). Assume that F is equicontinuous. For any 0-neighborhood $V \subset Y$, there exists a neighborhood U of the zero vector in X such that $F(U) \subset V$. For any $A \subset X$, there exists a number $r > 0$ such that $A \subset rU$, which implies that $F(A) \subset rV$ and thus $F(A)$ is bounded in Y. Therefore, F is equibounded.

Finally, we verify that if X, Y are B^* spaces, then (3) implies (1). Assume that F is equibounded. For any 0-neighborhood $V \subset Y$, there exists a number $r > 0$ such that $V \supset rB_Y$, where B_Y is the unit ball in Y. Since F is equibounded, there exists a neighborhood U of the zero vector in X such that $F(U) \subset rB_Y \subset V$. Therefore, F is equibounded at some neighborhood of the zero vector in X. □

Proposition (3.2) shows us the relationship between those three conditions. However, any of those three conditions are difficult to verify in general, while verifying that the condition

$$F(x) = \{Tx : T \in F\}$$

is bounded in Y for any $x \in X$ is much easier. The following is to prove that the boundedness of $F(x)$ for any $x \in X$ implies the equiboundedness of F under some mild conditions.

Theorem 3.7 (Uniform Boundedness Principle)**.** *Let X, Y be two TVSs and $F \subset \mathcal{L}(X, Y)$. If X is a Frechet space and for any $x \in X$, $F(x)$ is bounded in Y, then F is equibounded.*

Proof. Assume that $\|\cdot\|$ is the quasi norm on X and $B_r = \{x \in X : \|x\| < r\}$. We only need to verify that there exists a number $r > 0$ such that $F(B_r)$ is bounded in Y. Assume that for any $n \in \mathbb{N}$, $F(B_{1/n})$ is not bounded in Y. Then for any $n \in \mathbb{N}$, there exists a number $r_n > 0$ such that $F(B_{1/n}) \supset r_n B_Y$, where B_Y is the unit ball in Y. For any $n \in \mathbb{N}$, there exists a linear operator $T_n \in F$ such that $\|T_n\| > nr_n$, which implies that there exists a vector $x_n \in B_{1/n}$ such that $\|T_n x_n\| > nr_n > 0$. Thus we have

$$F(x_n) = \{Tx_n : T \in F\} \supset \{T_n x_n\}$$

is not bounded in Y, which contradicts the assumption. Therefore, there exists a number $r > 0$ such that $F(B_r)$ is bounded in Y, which implies that F is equibounded. □

It has a useful corollary as follows.

Corollary 3.2. *Let X, Y be two TVSs and $F \subset \mathcal{L}(X, Y)$. If X is a Frechet space and for any $x \in X$, $F(x)$ is bounded in Y, then F is equicontinuous. Suppose that X is a Banach space and Y is a B^* space, if $\sup_{T \in F} \|T\| = \infty$, then the set $E = \{x \in X : \sup_{T \in F} \|Tx\| < \infty\}$ is a first category set in X.*

Uniform boundedness principle is powerful when we analyze the convergence of a sequence of linear operators between two TVSs.

Theorem 3.8. *Suppose that $T_n \subset \mathcal{L}(X,Y)$ is a sequence of linear operators between two TVSs X, Y and denote*

$$A = x \in X : \lim_{n\to\infty} T_n x \text{ exists in } Y.$$

For any $x \in A$, we denote $T(x) = \lim_{n\to\infty} T_n x$.

(a) *If $A = X$ and X is a Frechet space, then $T \in \mathcal{L}(X,Y)$.*

(b) *If A is a second category set and Y is complete, then $T \in \mathcal{L}(X,Y)$.*

(c) *Suppose that $\overline{span(A)} = X$ and X, Y are Banach spaces. If $\sup_{n\in\mathbb{N}} \|T_n\| < \infty$, then $T \in \mathcal{L}(X,Y)$ and*

$$\|T\| \leq \varliminf_{n\to\infty} \|T_n\|.$$

Proof. (a) For any $x \in X$, we have $T_n x \to T(x)$, which implies that T is continuous at the zero vector and thus $T \in \mathcal{L}(X,Y)$.

(b) Since A is a second category set in X, there exists a neighborhood U of the zero vector in X such that $A \cap U$ is not a nowhere dense set in X. Thus there exists a neighborhood V of the zero vector in X such that $(A \cap U) + V \subset A \cap U$, which implies that for any $x \in V$, we have

$$T_n x = T_n(x + x_0) - T_n x_0 \to T(x + x_0) - T(x_0)$$

for some $x_0 \in A \cap U$. Thus we have that $T_n x \to T(x)$ for any $x \in V$, which implies that T is continuous at the zero vector and thus $T \in \mathcal{L}(X,Y)$.

(c) From (b) we know that $T \in \mathcal{L}(X,Y)$. Thus we have

$$\|Tx\| = \lim_{n\to\infty} \|T_n x\| = \varliminf_{n\to\infty} \|T_n x\| \leq \varliminf_{n\to\infty} \|T_n\| \cdot \|x\|$$

which implies that $\|T\| \leq \varliminf_{n\to\infty} \|T_n\|$.

□

The closed graph theorem and uniform boundedness principle are seemed to be the most powerful applications of the open mapping theorem. In fact, the following theorem shows that the open mapping theorem, closed graph theorem and uniform boundedness principle are equivalent.

Theorem 3.9. *The following conclusions are equivalent:*

(a) *Open mapping theorem: If X is a Frechet space and Y is a TVS, then any surjective linear operator $T \in \mathcal{L}(X, Y)$ is an open mapping.*

(b) *Closed graph theorem: If X, Y are two Frechet spaces and $T : X \to Y$ is a linear operator, then if the graph of T is closed in $X \times Y$, we have $T \in \mathcal{L}(X, Y)$.*

(c) *Uniform boundedness principle: If X, Y are two TVSs and $F \subset \mathcal{L}(X, Y)$, then if X is a Frechet space and for any $x \in X$, $F(x)$ is bounded in Y, we have that F is equibounded.*

Proof. We only need to verify that (1) implies (2), (2) implies (3) and (3) implies (1).

(1) $\Rightarrow$ (2): Assume that X, Y are two Frechet spaces and $T : X \to Y$ is a linear operator with a closed graph. We are going to verify that T is continuous. For any $y \in Y$ and $\epsilon > 0$, there exists a neighborhood V of the zero vector in Y such that $y + V \subset B(y, \epsilon)$. Since $G(T)$ is closed in $X \times Y$, we have that $G(T) \cap (X \times (y + V))$ is closed in $X \times (y + V)$. Since $G(T) \cap (X \times (y + V))$ is not empty, there exists a neighborhood U of the zero vector in X such that

$$(U \times (y + V)) \cap G(T) = (U \times (y + V)) \cap G(T) \cap (X \times (y + V)) = U \times (y + V).$$

Thus for any $x \in U$, we have

$$(x, T(x)) \in G(T) \cap (X \times (y + V)),$$

which implies that

$$T(x) \in y + V \subset B(y, \epsilon).$$

Therefore, T is continuous at the zero vector and thus continuous on X.

(2) $\Rightarrow$ (3): Assume that (2) holds. Let $F \subset \mathcal{L}(X, Y)$ be a family of linear operators from X to Y. We are going to verify that if for any $x \in X$, $F(x)$ is bounded in Y, then F is equibounded. For any finite subset $\{T_1, T_2, \cdots, T_n\}$ of F, we define a linear operator

$$T : X \to Y^n, x \mapsto (T_1 x, T_2 x, \cdots, T_n x).$$

Since $F(x)$ is bounded in Y for any $x \in X$, we have that T has a closed graph in $X \times Y^n$. From (2), we have that $T \in \mathcal{L}(X, Y^n)$, which implies that T is continuous and thus F is equibounded at some neighborhood of the zero vector in X. From Proposition (3.2), we have that F is equicontinuous and thus equibounded.

(3) $\Rightarrow$ (1): Assume that (3) holds. Let X be a Frechet space and Y be a TVS. We are going to verify that if $T \in \mathcal{L}(X, Y)$ is surjective, then T is an open mapping.

For any $r > 0$, we only need to verify that $T(B_r)$ is a neighborhood of the zero vector in Y, where $B_r = \{x \in X : \|x\| < r\}$. Assume that for any $n \in \mathbb{N}$, $T(B_{1/n})$ is not a neighborhood of the zero vector in Y. Then for any $n \in \mathbb{N}$, there exists a neighborhood V_n of the zero vector in Y such that $V_n \cap T(B_{1/n}) = \emptyset$. For any $n \in \mathbb{N}$, there exists a linear operator $T_n \in \mathcal{L}(X, Y)$ such that $T_n x = Tx$ for any $x \in B_{1/n}$ and $\|T_n\| > n$. Thus we have that $F = \{T_n : n \in \mathbb{N}\}$ is a family of linear operators from X to Y such that for any $x \in X$, $F(x)$ is bounded in Y, but F is not equibounded, which contradicts the assumption. Therefore, there exists a number $r > 0$ such that $T(B_r)$ is a neighborhood of the zero vector in Y, which implies that T is an open mapping. $\square$